

The problem: Reinventing

common capabilities

Introduction

From custom tools to built-in capabilities

Your journey:

Part 1

Understood tools fundamentally and built custom function tools

Part 2

Use production-ready tools provided by ADK

The shift:

Instead of writing implementation code, you import and use pre-built tools that ADK maintains.



The problem: Reinventing common capabilities

Let's say you want to add web search to your agent from part 1:

Python

```
# How would you implement web search?
def search_web(query: str) -> dict:
    """Search the web for information."""
    # How do we actually implement this?
    # - Need to call a search API (which one? how to authenticate?)
    # - Parse and format results for LLM consumption
    # - Handle rate limits and errors
    # - Keep API integration up to date
    # - Optimize results for the LLM
    # This is complex!
    pass
```

Issues with building everything yourself:

- ❌ **Complex implementation:** API integration, authentication, error handling
- ❌ **Maintenance burden:** API changes, updates, deprecations
- ❌ **Not optimized:** Results need to be formatted for LLM consumption
- ❌ **Missing production features:** Rate limiting, caching, monitoring
- ❌ **Time-consuming:** Hours or days to build what should take minutes

The root problem

Common capabilities like web search and code execution require significant engineering effort to implement well.